

# Novelty Assessment: Online Continual Learning in Decentralized Robot Agents for Autonomous Self-Healing (LeKiwi)

## TL;DR

**The specific combination is genuinely novel and unoccupied.** No paper from 2024–2026 unites an on-device small LLM whose weights are updated online (LoRA + replay) during deployment, applied specifically to sensor-fault compensation, coordinating decentralized subsystem-level (ARM/BASE/GRIPPER) health agents, on real mobile-manipulator hardware. The field bifurcates into "frozen LLM for fault recovery" vs. "online-learning VLA for general skills," and this thesis sits precisely in the gap between them.

**The single greatest novelty risk is the "online learning" claim, not the architecture.** The closest fault-recovery SLM work (Chen et al., Goal-oriented Communication, arXiv:2601.18765) uses offline LoRA and is simulation-only; the closest on-device Jetson continual-learning agent ("Forget to Improve," arXiv:2606.25115) explicitly keeps its planner frozen and only curates memory. The thesis must therefore prove that weight-level online adaptation beats memory/RAG and offline-trained baselines — that is where reviewers will push.

**Positioning should foreground the subsystem-health-agent decomposition and real-hardware online learning, which have no direct counterpart.** Adjacent decentralized fault-tolerance work is control-theoretic (Kalman/observer-based), and adjacent LLM-robot work is task-level; the "learning local LLM updating structured subsystem health agents" framing is the distinctive contribution.

## Key Findings

**1. The literature splits cleanly into two branches, and the thesis occupies the gap.** Branch A — LLM/SLM for robot fault recovery (REFLECT, DoReMi, RECOVER, AHA, Code-as-Monitor, Sinha et al. RSS 2024, STAR, IntelLiPlan, Chen et al. 2026) — keeps the LLM frozen at deployment. Branch B — on-device or continual learning on real robots — is either memory/RAG-based with frozen weights (Forget to Improve, arXiv:2606.25115, on Jetson Orin) or genuinely weight-updating but for general skill acquisition rather than fault compensation (RoboTTT arXiv:2607.15275; VLA continual learning arXiv:2605.26820). No work bridges them.

### 2. Closest prior art, ranked (from targeted search):

**Chen, Aijaz & Deng — "Goal-oriented Communication for Fast and Robust Robotic Fault Detection and Recovery"** (arXiv:2601.18765, 2026). Uses Llama-3.2-1B/11B LoRA-fine-tuned + knowledge distillation for recovery-motion generation. **But:** LoRA is offline; model frozen at deployment; MuJoCo simulation only; no on-device deployment. Closest on the "SLM + LoRA + fault recovery" axes.

**"Forget to Improve: On-Device LLM-Agent Continual Learning via Budget-Curated Memory"** (arXiv:2606.25115, 2026). Qwen2.5-7B on Jetson Orin/Thor, real embodied testbed. **But:** explicitly frozen planner — "not from changing the frozen planner"; continual learning = memory curation, exactly the mechanism the thesis differentiates from.

**RoboTTT** (arXiv:2607.15275, 2026). Test-time training with "fast weights... updated by gradient descent during both training and inference" on real robots, recovers from its own mistakes. **But:** general VLA policy, not fault compensation, not an SLM reasoning agent, not Jetson/on-device, no subsystem agents. **VLA continual learning with replay** (arXiv:2605.26820, 2026). Online fine-tuning of a VLA on sequential real-world data with experience replay buffers on a physical arm; "even small replay buffers substantially improving retention." **But:** skill acquisition, not fault compensation; VLA not on-device SLM; demonstration streams not autonomous.

**RDMM** (Nasrat, Kim, Lee, Lee, Jang & Yi, arXiv:2501.16899, 2025). On-device fine-tuned LLM for robot decision-making; the RDMM framework "can plan with a 93% accuracy" and is trained on "a new dataset consisting of 27k planning instances, as well as 1.3k text-image annotated samples," with RDMM-8B running at 5.54 tokens/s on a Xavier AGX 32GB and RDMM-0.5B at 3.75 tokens/s on an entry-level Xavier NX 8GB. **But:** offline fine-tuned, inference-only at deployment; no fault focus.

**3. Baselines are well-chosen and correctly characterized.** MROS (Bozhinoski et al., "MROS: runtime adaptation for robot control architectures," Advanced Robotics Vol. 36, No. 11, 2022; arXiv:2010.09145 / 2303.09227) is an ontology-based (TOMASys) implementation of the MAPE-K meta-control loop for rule-based architectural reconfiguration — a strong "no-learning" baseline, and notably already evaluated on a mobile-manipulator demonstrator: "an autonomous dual-arm mobile manipulator [ABB YuMi] building a pyramid." REFLECT (Zeyi Liu, Arpit Bahety & Shuran Song, CoRL 2023, arXiv:2306.15724), which "queries LLM for failure reasoning based on a hierarchical summary of robot past experiences generated from multisensory observations" and introduces the RoboFail dataset, and DoReMi (Guo et al., 2023) are the canonical frozen-cloud-LLM comparators. Miki, Lee, Hwangbo, Wellhausen, Koltun & Hutter, "Learning robust perceptive locomotion for quadrupedal robots in the wild," Science Robotics 7(62):eabk2822, 2022 — whose controller "was tested in a variety of challenging natural and urban environments over multiple seasons and completed an hour-long hike in the Alps in the time recommended for human hikers" — and the RMA line (Kumar et al. 2021) represent offline-trained fault-tolerance/adaptation policies. This baseline triad (rule-based / frozen-LLM / offline-RL) cleanly brackets the thesis's contribution.

## Details

**On-device LLM adaptation is feasible but under-demonstrated for learning.** LoRA/QLoRA fine-tuning of TinyLlama-1.1B and Mistral-7B on Jetson Orin is documented (NVIDIA Jetson AI Lab; NVIDIA Developer Forums report TinyLlama LoRA in ~22 min, 563k trainable params, 4.5GB GPU on Orin Nano 8GB), so the compute premise is sound. But nearly all robot deployments use these models for inference or offline-tuned adaptation, not online in-the-loop weight updates — reinforcing the novelty. [NVIDIA Developer](#)

[Forums](#)

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**Continual-learning machinery is mature and transferable.** The stability-plasticity/catastrophic-forgetting literature offers directly applicable tools: replay + LoRA hybrids (ERI-LoRA; I-LoRA dual fast/slow memory; SuRe surprise-driven replay; approximate regularized replay). Notably, phi-2 + LoRA continual fine-tuning with class-aware selective replay has already been demonstrated (Hybrid-CASR, arXiv:2602.23834) — for software vulnerability detection, not robotics, but it validates the exact Phi-2+LoRA+replay recipe the thesis proposes and is a useful methodological citation. [arxiv + 4](#)

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**Decentralized fault detection exists but is control-theoretic, not learning-based.** Multi-robot/multi-subsystem FDI work (e.g., decentralized LiDAR-fault FTC, Sci. Reports 2024; swarm fault detection, PLOS ONE 2017; hierarchical spacecraft FDI) decomposes systems into local diagnosers with a supervisor — structurally analogous to ARM/BASE/GRIPPER agents + LLM coordinator — but uses Kalman filters/observers/thresholds, with no LLM and no learning. Sensor-fault compensation on manipulators (adaptive observers, sliding-mode, active inference precision-learning) is likewise analytic. The thesis's "structured health reports consumed by a learning LLM coordinator" is a genuinely different abstraction.

**LLM-robot health/diagnosis agents are emerging but frozen and task-level.** ROSA (NASA-JPL) and ROS Help Desk provide LLM-based ROS diagnosis; PHM-Society work uses LLM agents for fault diagnosis from raw sensor data. All are frozen-inference, and mostly diagnostic/interfaces rather than closed-loop self-healing with learning. [GitHub + 2](#)

**Platform grounding is solid.** LeKiwi (SIGRobotics-UIUC) is a real, low-cost open-source omnidirectional mobile manipulator: per the Sseed Studio/OpenELAB LeKiwi Kit spec it uses "Feetech STS3215 servos (30kg torque per joint)... Omnidirectional drive... compatible with Raspberry Pi 5, enabling seamless deployment of the LeRobot framework," with a 6-axis SO-ARM101/100 and a three-wheel kiwi drive (omniwheels at 120° for holonomic motion). It supports Raspberry Pi 5 / Jetson Orin NX compute — making a ~50-trial handover/fetch-place hardware validation credible and reproducible. [OpenELAB](#)

## The 10-Line Senior Researcher's Assessment

**Verdict: the combination is genuinely novel.** No 2024–2026 paper unites on-device online-learning SLM + subsystem health agents + sensor-fault compensation + real mobile-manipulator hardware; the decentralized ARM/BASE/GRIPPER health-agent framing has no counterpart.

**The load-bearing novelty is "learning at deployment," not the LLM-coordinator idea** — LLM fault reasoning (REFLECT/DoReMi/RECOVER/Sinha) and on-device LLMs (RDMM) already exist, all frozen.

**Closest single threat is Chen et al. 2026 (arXiv:2601.18765)** — SLM+LoRA for fault recovery — but it is offline-trained and simulation-only, so it is distinguishable if you demonstrate online adaptation on hardware.

**Second watch-item is "Forget to Improve" (arXiv:2606.25115)** — on-device Jetson continual learning — which deliberately freezes weights and curates memory; you must argue and empirically show why weight-level LoRA beats memory/RAG.

**Your baseline triad (MROS rule-based / frozen cloud LLM / offline-RL à la Miki-RMA) is exactly right** and brackets the contribution cleanly; keep all three.

**Strategic positioning: lead with two claims** — (a) subsystem-level (not task-level) health agents, and (b) online weight-level self-improvement on real hardware; these are the defensible, quotable deltas.

**Biggest scientific risk is the stability-plasticity tradeoff on a tiny model** — Phi-2/TinyLlama online LoRA can forget base reasoning; adopt dual fast/slow-LoRA + prioritized replay (I-LoRA/SuRe/ERI-LoRA) and measure forgetting explicitly.

**Biggest methodological risk is ~50 trials** — too few for a strong learning-curve claim; add controlled fault-injection episodes and report retention/transfer metrics, not just success rate.

**Cite the phi-2+LoRA+selective-replay precedent (arXiv:2602.23834)** to pre-empt "is on-device continual LoRA even feasible" objections, and Jetson LoRA timing benchmarks to prove the compute budget.

**Direction clarity is high; scope discipline is the main threat** — resist adding VLA control learning; keep the thesis about the LLM coordinator learning fault-compensation *strategies* over frozen low-level skills, which is what makes it unique and tractable.

## Recommendations

**Stage 1 — Lock the novelty claim now (months 0–2).** Write the related-work chapter around the two-branch bifurcation above. Explicitly position against Chen et al. 2026, "Forget to Improve," RoboTTT, and the VLA-replay paper, using the frozen-vs-learning and task-vs-subsystem axes. Benchmark to change this recommendation: if a new preprint appears that does online weight updates for robot *fault* compensation on hardware, narrow your claim to the subsystem-health-agent decomposition + human-handover domain.

**Stage 2 — De-risk the learning core before hardware (months 2–6).** Prototype Phi-2/TinyLlama online LoRA with dual-memory replay on logged/synthetic fault episodes; establish forgetting and plasticity curves offline. Threshold: if online LoRA cannot retain base reasoning while acquiring  $\geq 2$  sequential fault-compensation strategies without a replay buffer, adopt I-LoRA/SuRe-style dual fast/slow adaptation before touching hardware.

**Stage 3 — Hardware validation with fault injection (months 6–12).** Run the LeKiwi handover/fetch-place trials, but augment the ~50 trials with a structured fault-injection protocol (encoder bias, camera misalignment, gripper force dropout) so each fault class has enough repetitions to show a learning curve. Report against all three baselines. Threshold: an online-learning win is only credible if it beats the frozen-LLM baseline on *previously unseen* fault instances AND retains earlier fault competence (measured retention  $\geq$  baseline).

**Stage 4 — Ablations that reviewers will demand.** (a) memory/RAG vs. LoRA weight update (directly rebuts "Forget to Improve"); (b) centralized single-agent vs. decentralized subsystem agents; (c) replay-buffer on/off. These three ablations convert the novelty claim into a defended contribution.

## Caveats

Several of the closest papers are very recent 2026 arXiv preprints (IDs in the 2601–2607 range) read partly via abstract/intro; the frozen-vs-learning classifications rest on explicit verbatim quotes (high confidence) but warrant full-paper verification before the final thesis, as the 2026 preprint frontier is moving fast and a direct competitor could appear during the PhD.

"Novel" here means the *combination* is unoccupied; every individual ingredient (LoRA on Jetson, replay-based CL, LLM fault reasoning, decentralized FDI, LeKiwi) is well established — the contribution is integrative and empirical, so the thesis must win on demonstrated hardware results, not architectural claims alone.

Some retrieved sources are vendor/blog material (NVIDIA Jetson AI Lab, developer forums, product pages); these support feasibility/compute claims only and should not be cited as peer-reviewed evidence. The Sakana "Text-to-LoRA/Doc-to-LoRA" and SEAL self-adapting-LM lines suggest rapid instant-adapter alternatives to gradient-based online LoRA; worth monitoring as either a threat or an alternative mechanism for the thesis.

Note on platform naming: the task refers to "LeKwik," but the actual open-source platform is **LeKiwi** (SIGRobotics-UIUC); this assessment assumes they are the same target robot.